Spring, 2023

Marmara University
Engineering faculty
Electrical and Electronics Engineering Department
EE4077 Fundamentals of Machine Learning
Final Exam

1. Consider the following design matrix, representing four sample points X_i in $\mathbb{R}^2$. We want to represent the data in only one dimension, so we turn to principal components analysis (PCA).

$$X = \begin{bmatrix} 4 & 1 \\ 2 & 3 \\ 5 & 4 \\ 1 & 0 \end{bmatrix}$$

(a) Compute the unit-length principal component directions of X, and state which one the PCA algorithm would choose if you requested just one principal component. Please provide an exact answer, without approximation. Show your work! [10 pt]

(b) The plot on the left below depicts the sample points from X. We want a one-dimensional representation of the data, so draw the principal component direction (as a line) and projections of all four sample points onto the principal direction. Label each projected point with its principal coordinate value. Give the principal coordinate values exactly. [10 pt]

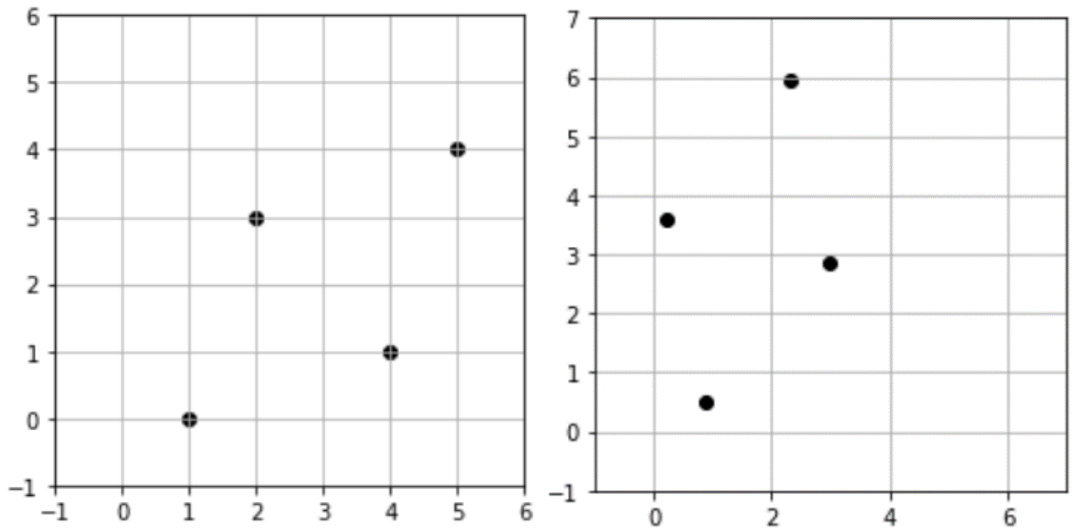


Figure: (left) original samples, (right) original samples rotated 30 degrees counterclockwise

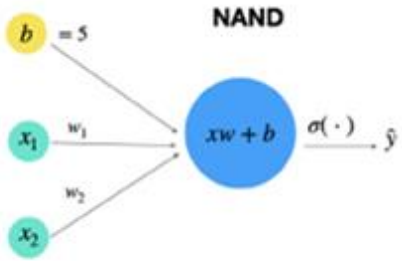
(c) The plot on the right above depicts the sample points from X rotated 30 degrees counterclockwise about the origin. Identify the principal component direction that the PCA algorithm would choose and draw it (as a line) on the plot. Also draw the projections of the rotated points onto the principal direction. Label each projected point with the exact value of its principal coordinate. [5 pt]

2.

(a) Consider a single unit in a neural network that receives two binary inputs $x_1, x_2 \in \{0,1\}^2$ and computes a linear combination followed by a threshold activation function, namely,

$$\sigma(z) = \begin{cases} 1, & z \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

The unit is illustrated below. We have chosen a bias term of $b = 5$. **Provide values for the two weights w_1 and w_2 that allow you to compute the NAND function** (which is 0 if and only if both inputs are 1). [10 pt]



(b) We are drawing sample points from a distribution with the probability density function (PDF) $f(x) = \frac{1}{2} e^{-|x-\mu|}$, but we do not know the mean $\mu \in \mathbb{R}$. We decide to estimate μ with maximum likelihood estimation (MLE). Unfortunately, we have only two sample points $X_1, X_2 \in \mathbb{R}$.

Derive the likelihood and the log-likelihood for this problem. Then show that every value of μ between X_1 and X_2 is a maximum likelihood estimate [15 pt]

3. Suppose that we want to classify images that belong to one of ten classes (i.e. [cat, dog, bird, turtle, ..., horse]). The images come in RGB format (one channel for each color) and are down sampled to dimension 128x128. We construct a Convolutional Neural Network that has the following structure: the input is first max pooled with a 2x2 filter with stride 2 and 3 output channels. The results are then sent to a convolutional layer that uses a 17x17 filter of stride 1 and 12 output channels. Those values are then passed through a max pool with a 3x3 filter with stride 3 and 12 output channels. The result is then flattened and passed through a fully connected layer (ReLU activation) with 128 hidden units followed by a fully connected layer (SoftMax activation) with 10 hidden units. We say that the final 10 hidden units thus represent the categorical probability for each of the ten classes. With enough labeled data, we can simply use some optimizer like SGD to train this model through backpropagation. (Please assume we have bias terms in all neural network layers unless explicitly stated otherwise).

   (a) Draw a diagram that illustrates the channels and dimensions of the tensors before and after every neural net operation [10 pt]
   (b) How many parameters are in this network for the convolutional components? [5 pt]
   (c) How many parameters are in this network for the fully connected components? [5 pt]
   (d) From these parameter calculations, what can you say about convolutional layers and fully connected layers in terms of parameter efficiency? Why do you think this is the case? [5 pt]

4 For True (T) or False (F) questions, justify your answer, for others write down your answer.

   (a) (T or F) For a particular dataset and a particular k, k-means always produce the same result if the initialized centers are the same. Assume there is no tie when assigning the clusters. [5 pt]]
   (b) (T or F) k-means can always converge to the global optimum. [5 pt]
   (c) (T or F) k-means is not sensitive to outliers. [5 pt]
   (d) k in k-nearest neighbors and k-means have the same meaning. [5 pt]
   (e) What is the biggest difference between k-nearest neighbors and k-means? [5 pt]

T=k-means is deterministic when the initial centers are fixed and there is no tie in assigning points to clusters.
F=k-means often converges to a local optimum, not necessarily the global optimum.
F=k-means is sensitive to outliers because they can distort the means of clusters
F=k-nearest neighbors (k-NN), k represents the number of nearest neighbors considered for classification or regression. In k-means, k represents the number of clusters to partition the data into.

e)The biggest difference is that k-nearest neighbors is a supervised learning algorithm used for classification and regression, while k-means is an unsupervised learning algorithm used for clustering.